

What are your thoughts about advanced data analytics tools and optimization methods in general?

Advanced data analytics tools help us find useful information from data to make better decisions. They are interesting because they solve real problems and use skills like math and coding. However, they can be tough because you need to understand complex ideas and clean messy data. Despite the difficulty, they are rewarding and exciting because they help improve things in the real world.

What is easy in advanced data analytics and data optimization, what is difficult or complex?

Advanced data analytics and optimization can be easy when using tools like **Python, Pandas, NumPy**, and visualization libraries such as **Matplotlib** and **Seaborn**, which were used in the assignments to explore, clean, and visualize data. Data cleaning using **Pandas functions** (handling missing values, filtering, and transformations) feels manageable with practice. However, the difficult part is advanced data cleaning, feature engineering, and optimization, especially when using **scikit-learn** for model building, preprocessing, and tuning, because these steps require deeper understanding, careful decisions, and experience. Overall, using the tools is easy, but preparing data correctly and choosing the right analytical or optimization approach is the most complex part.

What would you like to learn next in machine learning?

I would like to learn more about **real-world machine learning and AI applications**, especially working with **large and messy datasets**. I am interested in building **AI-powered systems** that can make predictions and recommendations. In the future, I would like to code machine learning models using **scikit-learn and AI techniques** and integrate them into **mobile applications**, where users can automatically interact with others who have similar interests. This would help create smarter, more personalized, and user-focused applications.

Can you use your advanced data analytics skills in your current activities and/or hobbies? Or even in your work? Would you like to work as a professional machine learning engineer in the future?

Yes, I can use advanced data analytics skills in my current work because I am working in a company and planning to start my thesis based on it. Advanced analytics can help analyze company data, improve decision-making, and build intelligent systems. For example, I can develop an **AI-based application where users automatically interact with others who have similar interests** by using **machine learning techniques such as clustering, recommendation systems, and natural language processing (NLP)**. User data and text inputs can be analyzed to group similar users and suggest connections. A **chatbot** powered by **AI and NLP** can also be integrated to guide users, answer questions, and improve user interaction. Yes, I would like to work as a **professional machine learning engineer in the future** because it allows me to apply advanced data analytics and AI techniques to solve real-world problems and build smart, user-centered applications.

How do you understand the following concepts and/or technologies? In your own words, write about what you can do with them while performing advanced data analytics, and where they are useful.

Dataset evaluation tools (Ydata-profiling, Sweetviz, dataprep etc.)

Dataset evaluation tools like **YData Profiling**, **Sweetviz**, and **Dataprep** make data analysis easier by quickly showing data quality, patterns, missing values, and relationships, helping users understand datasets before modeling.

Distribution management (e.g., noise/outlier/overlap/redundancy management, data synthesis, sampling techniques etc.)

Distribution management in data involves handling **noise, outliers, overlaps, and redundancy**, creating synthetic data, and using **sampling techniques** to ensure datasets are clean, balanced, and suitable for accurate modeling and acceptable for even models.

Feature engineering tools (optimal variable selection, feature generation, high cardinality management methods, etc.)

Feature engineering tools help select the most relevant variables to improve model performance. They generate new features from existing data and manage high-cardinality variables effectively. These tools reduce noise, improve accuracy, and make models more efficient and interpretable.

The role of advanced dataset optimization methods in machine learning applications.

Advanced dataset optimization improves data quality by reducing noise and redundancy. It enhances model accuracy, efficiency, and overall performance in machine learning applications.

The perfect amount of optimization balances the dataset to avoid **underfitting** (model too simple to learn patterns) and **overfitting** (model too tuned to training data). It ensures the model learns meaningful patterns while generalizing well to new, unseen data.

If you participated in the semester project this semester, describe your personal advanced data analytics–related tasks in the project.

As of now, I am working in a company, analyzing data to understand user interests and creating connections based on those interests. In this semester project, I am part of a team where we collect data from AI, filter it according to users' best interests, and display results based on the best match. It is a chatbot designed specifically for users to find exact information while avoiding unnecessary context.

Any other feedback considering the course itself or advanced data analytics / optimization in general? Both positive and negative feedback is welcome!

The course provides a strong foundation in advanced data analytics techniques, including feature engineering, dataset optimization, and AI-driven solutions. Practical projects and hands-on assignments help in understanding real-world applications.

Generally

I really liked the teaching style and methods in this course. Most topics were taught with hands-on examples, live coding, and useful templates, which made learning much easier. Even though the concepts were tough, the clear explanations and materials helped a lot.

